

YACHAY TECH UNIVERSITY

SYLLABUS

1. General Information				
A. School	School of Physical Sciences and Nanotechnology	B. Major	Physics	
C. Course	Special Topics in Computational Physics	D. Code	ECFN-PHY-1112	
E. Level	Not Applicable	F. Academic Term	Second Semester 2026	
G. Curricular Unit	Professional	H. Study Mode	In person	
I. Total Hours	144	J. Professors	Wladimir Eduardo Banda Barragán	
K. Weekly Class Schedule	17:00 - 19:00 Tuesday 17:00 - 19:00 Wednesday 13:00 - 15:00 Thursday	L. Weekly Tutoring Schedule	16:00 - 17:00 Tuesday 16:00 - 17:00 Wednesday	

2. Prerequisites and Corequisites			
Prerequisites		Corequisites	
Course	Code	Course	Code

3. Course Description

This course develops computational fluid dynamics as a current research topic in computational physics, using specialised computing, simulation and visualisation tools throughout. Starting from the fundamental properties of fluids, hydrostatics and control-volume analysis, it builds the differential formulation of fluid motion and the numerical methods used to solve it. Topics range from flow kinematics, transport equations and numerical stability, through the Navier–Stokes equations, pressure–velocity coupling and two-dimensional incompressible solvers, dimensional analysis and similitude, solver verification and validation, and laminar and turbulent internal and external flow with boundary layers, lift and drag, to compressible flow with shocks, hydrodynamic instabilities, parallel computing and mesh-free particle methods. Each unit combines the underlying theory with the design, implementation and validation of numerical solvers, and with the analysis and visualisation of the resulting simulation data.

4. Learning Outcomes of the Course

- Apply the fundamental principles of fluid mechanics to analyse fluids at rest and in motion, and to formulate the governing conservation laws in integral and differential form.
- Select and implement numerical methods appropriate to each flow regime, including high-resolution finite-volume schemes for compressible flow and projection methods for incompressible flow.
- Use specialised computing, simulation and visualisation tools to design, verify and validate solvers for laminar and turbulent flows, assessing accuracy, stability and convergence against analytical and experimental benchmarks.
- Analyse and visualise simulation data to characterise turbulence, boundary layers, hydrodynamic instabilities, and aerodynamic forces in engineering and physical systems.



5. Units / Contents / Hours / Evaluation Instruments and Schedule						
Curricular Units	Contents	Teaching Hours	Hours of Internship and Experimental Learning		Hours of Independent Learning	Evaluation Instruments and Schedule
			Hours in Contact	Hours not in Contact		
Unit 1. Fluids, Conservation Laws and a First Solver	Fluid properties, the continuum hypothesis, viscosity, and surface tension	2	2	0	2	In-class quizzes and project submissions with defences, with dates shared in a timely manner by the lecturer. Midterm exam (September 30 to October 8). Final exam (December 7 to 10).
	Fluid statics: pressure distribution, manometry, hydrostatic forces, and buoyancy	2	2	0	2	
	Control-volume analysis: conservation of mass, momentum and energy; Bernoulli's equation	2	2	0	2	
	From conservation laws to computation: grids, discretisation, and a first advection solver	2	2	0	2	
Unit 2. Kinematics, Transport and Numerical Stability	Flow kinematics: Lagrangian and Eulerian descriptions, material derivative, streamlines and pathlines	2	2	0	2	In-class quizzes and project submissions with defences, with dates shared in a timely manner by the lecturer. Midterm exam (September 30 to October 8). Final exam (December 7 to 10).
	Linear and nonlinear transport: wave steepening and shock formation in Burgers' equation	2	2	0	2	
	Diffusion, the CFL condition and numerical stability: numerical dissipation and dispersion	2	2	0	2	
	Finite-volume formulation: conservative fluxes, boundary conditions, and verification tests	2	2	0	2	
Unit 3. Differential Analysis and Viscous Flow	Differential analysis: continuity and Navier–Stokes equations, stream function, and vorticity	2	2	0	2	In-class quizzes and project submissions with defences, with dates shared in a timely manner by the lecturer. Midterm exam (September 30 to October 8). Final exam (December 7 to 10).
	Exact solutions and potential flow: Couette and Poiseuille flow, and elementary flow superposition	2	2	0	2	
	Advection and diffusion in two dimensions: the vorticity–streamfunction formulation	2	2	0	2	
	Pressure–velocity coupling: the pressure–Poisson equation and projection methods	2	2	0	2	

Curricular Units	Contents	Teaching Hours	Hours of Internship and Experimental Learning		Hours of Independent Learning	Evaluation Instruments and Schedule
			Hours in Contact	Hours not in Contact		
Unit 4. Incompressible Solvers, Similitude and Validation	Two-dimensional incompressible solvers: lid-driven cavity and channel flow	2	2	0	2	In-class quizzes and project submissions with defences, with dates shared in a timely manner by the lecturer. Midterm exam (September 30 to October 8). Final exam (December 7 to 10).
	Flow past a cylinder: vortex shedding, Strouhal number, and wake structure	2	2	0	2	
	Dimensional analysis, the Buckingham Pi theorem, similitude, and dimensionless groups	2	2	0	2	
	Verification and validation: grid convergence, benchmark data, and open-source flow solvers	2	2	0	2	
Unit 5. Internal and External Flow, Boundary Layers and Turbulence	Laminar and turbulent internal flow: velocity profiles, friction factors, and the Moody chart	2	2	0	2	In-class quizzes and project submissions with defences, with dates shared in a timely manner by the lecturer. Midterm exam (September 30 to October 8). Final exam (December 7 to 10).
	Boundary layers: displacement and momentum thickness, transition, and flow separation	2	2	0	2	
	Drag and lift on immersed bodies: pressure and friction contributions, and aerofoil polars	2	2	0	2	
	Turbulence: the energy cascade, statistical description, and turbulence modelling in simulations	2	2	0	2	
Unit 6. Compressible Flow, Instabilities and Large-Scale Simulation	Compressible flow: speed of sound, Mach number, isentropic relations, and nozzle flow	2	2	0	2	In-class quizzes and project submissions with defences, with dates shared in a timely manner by the lecturer. Midterm exam (September 30 to October 8). Final exam (December 7 to 10).
	Shocks and expansion waves: the Riemann problem and Godunov-type finite-volume schemes	2	2	0	2	
	High-resolution methods and hydrodynamic instabilities: Kelvin–Helmholtz, Rayleigh–Taylor, Richtmyer–Meshkov	2	2	0	2	
	Beyond grid-based serial solvers: parallel domain decomposition and mesh-free particle methods	2	2	0	2	
Total		48	48	0	48	144

6. Teaching Methodology

Learning Environments

Learning Environment	Characteristics
Real: Digital or printed materials designed by the lecturer.	Real: Classroom equipped with a white board, which promotes the development of communication skills through interactive in-person activities. Access to a laboratory with computers and specific software for programming, data analysis, and simulation activities.
Virtual: MOODLE platform, GitHub, the Internet, websites, theoretical classes via ZOOM or Google Meet, seminars/workshops, tutoring, and other technological resources.	Virtual: Educational platform for sharing presentations, homework, code, assessments, and student participation forums.
Classroom: Theoretical classes, seminars/workshops, exercise classes, tutoring.	

Learning Tools

Learning Tools	Characteristics
Interactive lectures include theory and worked examples, in-class/lab assignments include simulation exercises and quizzes on reading material, and individual and group projects include computational modelling and research assignments.	These tools help students develop computing, simulation and visualisation skills for modelling fluid flows in the physical sciences.



7. Information Sources (Bibliography)

7.1 Main

Author/s	Title of Work	Edition	Year of Publication	Publishing house - Country
Çengel, Yunus A.; Cimbala, John M.	Fluid mechanics: fundamentals and applications	3rd	2014	McGraw Hill - United States
White, Frank M.	Fluid mechanics	7th	2011	McGraw Hill - United States

7.2 Complementary

Author/s	Title of Work	Edition	Year of Publication	Publishing house - Country
Munson, Bruce R.; Okiishi, Theodore H.; Huebsch, Wade W.; Rothmayer, Alric P.	Fundamentals of fluid mechanics	7th	2013	Wiley - United States
Toro, Eleuterio F.	Riemann solvers and numerical methods for fluid dynamics: a practical introduction	3rd	2009	Springer - Germany



8. Student's Evaluation

8.1 First Term of the Period*

Formative Evaluation		Laboratory**		Midterm Evaluation		Total
Project average (code routines and homework defences)	12.5%	Quiz average (physics, math and programming exercises)	12.5%	1 Midterm Exam	25%	50 %
Subtotal	12.5%	Subtotal	12.5%	Subtotal	25%	

8.2 Second Term of the Period

Formative Evaluation		Laboratory**		Final Evaluation		Total
Project average (code routines and homework defences)	12.5%	Quiz average (physics, math and programming exercises)	12.5%	1 Final Exam	25%	50 %
Subtotal	12.5%	Subtotal	12.5%	Subtotal	25%	

Evaluation Considerations

Make Up Exam

According to Article 61 of the UITEY Academic Regulations, "if students have a final grade point average of 4.8 to 5.9 in the subject, they will be entitled to a make-up exam at the end of the term. The content of this exam must reflect all aspects covered in the subject during the academic term. (...).

A passing grade on a make-up exam will raise the student's total grade to 6.0 (the minimum passing grade)."

Student Attendance to Academic Activities

Students must comply with Articles 82 and 83 of the UITEY Academic Regulations, which state:

"Article 82.- Student Attendance at Academic Activities. – Students must attend all academic activities, both synchronous and asynchronous, such as classes, assessments, laboratory practices, and other activities that contribute to their overall education, on the established dates and times. Academic staff must ensure compliance with the attendance criteria and the learning components in contact with the faculty and experimental practice through attendance records in accordance with the mechanisms defined by the Academic Vice-Rectorate.

For second language learning subjects, a minimum attendance of eighty percent (80%) is required for passing. The minimum attendance required for passing the other subjects will be seventy percent (70%).

Article 83.- Justification for absence. - Students who are absent must present supporting documents within a maximum period of five (5) days, counting from the date of absence.

The justifications will be validated by the teacher or the competent unit. Once the justification has been approved, the teacher, in his or her discretion, may define the criteria for making up activities missed due to non-attendance."

(*) Teaching staff will record the scores generated up to the first term of the period in Moodle, by the deadline established in the academic calendar. To determine the weights of each item, the provisions of Article 63 of the UITEY Academic Regulations must be observed.

(**) For courses without a laboratory component, indicate: N/A.

9. General Considerations

- Students are responsible for ensuring the academic integrity of their submitted assignments and exams.
- Cheating in exams, plagiarising, and copying solutions from other students, from chatbots automated by artificial intelligence, from solution manuals, or from previous years' solutions are all breaches of academic integrity.
- Academic misconduct will be penalised according to the University's regulations.
- Assignment deadlines and exam dates will be discussed and agreed upon in class. Once fixed, they are hard deadlines.

Prepared by	Reviewed by	Approved by
Professor - Professors	Designated Personnel	Dean – Language Director
Signature and Date:  Validar únicamente en FirmaBC. Firmado electrónicamente por: VLADIMIR EDUARDO SANDA BARRAGAN	Signature and Date:	Signature and Date: